

Operating-Point versus Capability: A Preregistered, Blind, Deterministic Reproducibility Study of Inference-Time Mechanisms in ARC-Style Reasoning

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Abstract

When an inference-time mechanism improves a benchmark score, is the gain *capability*, or a change in the coverage/confident-error operating point? We report a preregistered, adversarially-audited program (62 recorded claims, 34 documented failures, 46 experiments) that separates capability from answer-selection, abstention, verification, search, and evaluation policy using controlled deterministic experiments and a symbolic-solver invariance baseline. Across three open-weight families (Qwen2.5-7B, Mistral-7B, Phi-3.5-mini; $N=473$ each) and a 120-task ARC transfer subset, a diverse-prompt abstention mechanism reduces confident-wrong errors primarily by abstaining (reducing coverage), with no net capability gain (mechanism all-task accuracy $\leq$ baseline everywhere). A purported composition-depth “capability wall” is *search-limited*: it recovers fully as search depth increases at fixed capability. Benchmark score is *not invariant* to construct-irrelevant presentation and evaluation-policy changes (selection policy alone moves accuracy ~ 60 pp at fixed capability), while a capability-grounded symbolic solver is exactly invariant. An apparent $\leq 9\text{B} \rightarrow 14\text{B}$ parity “transition” *did not reproduce* under a fresh balanced forced-answer comparison (7B 0.80 $\approx$ 14B 0.825). We claim no novel mechanism; every effect maps to known theory. The contribution is a reproducible negative synthesis plus benchmark-design lessons, released as a public, CPU-only, standard-library replication package. External independent replication is *pending*.

1 Introduction

Reported gains from inference-time methods are often ambiguous between genuine capability and operating-point shifts. We distinguish six variables and hold candidate generation fixed while varying selection, abstention, verification, search, and evaluation policy. The work reads as evaluation/reproducibility research; it does not establish any universal or AGI capability claim.

2 Related Work

Selective prediction and the reject option (El-Yaniv & Wiener, 2010); capability versus skill (Chollet, 2019); library learning / search (Ellis et al., 2020); emergence-as-artifact (Schaeffer et al., 2023); prompt/format sensitivity (Sclar et al., 2023). All mechanisms here are **KNOWN**; our contribution is a controlled deterministic demonstration and honest negative synthesis, not a new mechanism.

3 Methods

Preregistered A/B: A = greedy always-answer; B = three diverse prompts, unanimous-or-abstain. We separate *candidate generation* from *candidate selection*. Metrics: accuracy, coverage, selective accuracy, confident-wrong. Statistics: Wilson 95% CIs, paired McNemar, Fisher exact, TOST (± 5 pp). Seeds {1234, 2025, 7}. Benchmark locked (SHA256 78446ba9..., 473 tasks). Open-weight models on free T4/P100.

4 Formal Model

Let C (capability), A (answer-selection), V (verification), Q (confidence quality), R (search), Y (correctness). C is latent; A, V, R, Y are observed under intervention. The LLM-side decomposition is *partially identified*: we can bound the selection/abstention contribution but do not claim full causal identification of C .

5 Results

Three families. Confident-wrong falls (Qwen 182→4, Mistral 252→19, Phi 124→9) via abstention; B all-task accuracy $\leq A$ in all three (no net capability gain). **ARC transfer.** A 0/120; B abstains ~99%, 0 solved. **Search-limited boundary.** Composition-depth failure recovers as search depth rises (M4 0.033 → 1.0): **BOUNDARY_CONFOUNDED (search-limited)**, not a capability wall. **Evaluation (non)invariance.** Symbolic solver exactly invariant ($I_{\text{task}}=1.0$); LLM score shifts ~60pp (selection) at fixed capability. **Scale (corrected).** Fresh balanced forced-answer: 7B 0.80 $\approx$ 14B 0.825 — the earlier gap did not reproduce (**TRANSITION_CONFOUNDED + KNOWN_EFFECT**); 14B does recognize parity on held-out data (even 0.92 / odd 0.78, Fisher $p=4.6\text{e}-13$), a narrow base-capability effect, not general reasoning.

6 Limitations

Synthetic/micro-world tasks; three families; $\leq 14\text{B}$; benchmark contamination concern (parity likely pre-trained); distribution-shift unresolved; R1 partial-by-design blindness; external replication pending; ARC transfer not justified; no universal AGI capability claim is established.

7 Reproducibility Statement

Public repository (anonymized for review) at a pinned commit; benchmark SHA256 78446ba9... (473 tasks); R1/R2/R3 runners use the Python standard library only (CPU, deterministic, no network/API); measured runtimes R1 0.02s, R2 0.10s, R3 0.06s; seeds {1234, 2025, 7}. Original operator outputs are preserved under a separate directory; R2/R3 replication paths are blind; R1 is partial-by-design; external replication remains pending.

8 Conclusion

Inference-time mechanisms shift the operating point (selection/abstention/search), not capability, across three families and ARC; benchmark scores are not invariant to construct-irrelevant conditions. Key lesson: answer-balance evaluations and report under fixed policy, validated presentation, and matched compute.

References

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